TELECOM CUSTOMER CHURN MODEL

1. Introduction

Customer churn occurs when a customer stops using the services of a company. In the telecom industry, predicting churn can help identify customers who are likely to leave. In this project, a Machine Learning model is developed using the Random Forest algorithm to predict customer churn based on customer and service-related information.

2. Objective

The main objective of this project is to develop a Machine Learning model that can predict whether a telecom customer is likely to churn or continue using the company's services.

- To analyze telecom customer data.
- To preprocess categorical and numerical customer information.
- To train a Machine Learning classification model.
- To classify customers into churn and non-churn categories.
- To evaluate the performance of the trained model using different classification metrics.

3. Dataset & Preprocessing

The IBM Telco Customer Churn dataset is used in this project. It contains **7,043 customer records and 21 columns**, including customer details, services, contract information, charges and churn status.

For preprocessing, the customerID column is removed, TotalCharges is converted into numerical format and missing values are filled using the median. Categorical attributes are converted into numerical values using Label Encoding. Finally, the dataset is divided into **80% training data and 20% testing data**.

4. Code Implementation

The project is implemented in Python using Pandas, Matplotlib and Scikit-learn. The complete implementation is given below:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, ConfusionMatrixDisplay

url = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.
data = pd.read_csv(url)

print(data.head())
print("Dataset shape:", data.shape)

data.drop("customerID", axis=1, inplace=True)
data["TotalCharges"] = pd.to_numeric(data["TotalCharges"], errors="coerce")
```

```
data["TotalCharges"] = data["TotalCharges"].fillna(data["TotalCharges"].median())

le = LabelEncoder()
for col in data.select_dtypes(include=["object", "str"]).columns:
    data[col] = le.fit_transform(data[col])

X = data.drop("Churn", axis=1)
y = data["Churn"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

prediction = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, prediction))
print("\nClassification Report:")
print(classification_report(y_test, prediction))

cm = confusion_matrix(y_test, prediction)
display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["No Churn", "Churn"]
)
display.plot()
plt.title("Confusion Matrix")
plt.show()
```

Output

Dataset Information	Value
Total Customer Records	7,043
Total Columns	21
Training Data	80%
Testing Data	20%
Testing Records	1,409

Performance Metric	Result
Accuracy	**79.56%**

Classification Report

Class	Precision	Recall	F1-Score	Support
No Churn (0)	0.83	0.91	0.87	1036
Churn (1)	0.66	0.47	0.55	373
Macro Average	0.74	0.69	0.71	1409
Weighted Average	0.78	0.80	0.78	1409

Overall Accuracy: 79.56%

The output shows that the model performs better in identifying customers who do not churn, with a recall of 0.91 for the No Churn class. For customers who churn, the model achieved a precision of 0.66 and recall of 0.47.

5. Machine Learning Model

The **Random Forest Classifier** is used for churn prediction. It combines multiple decision trees to generate the final classification result. The model uses **100 decision trees** and is trained on 80% of the dataset, while the remaining 20% is used for testing.

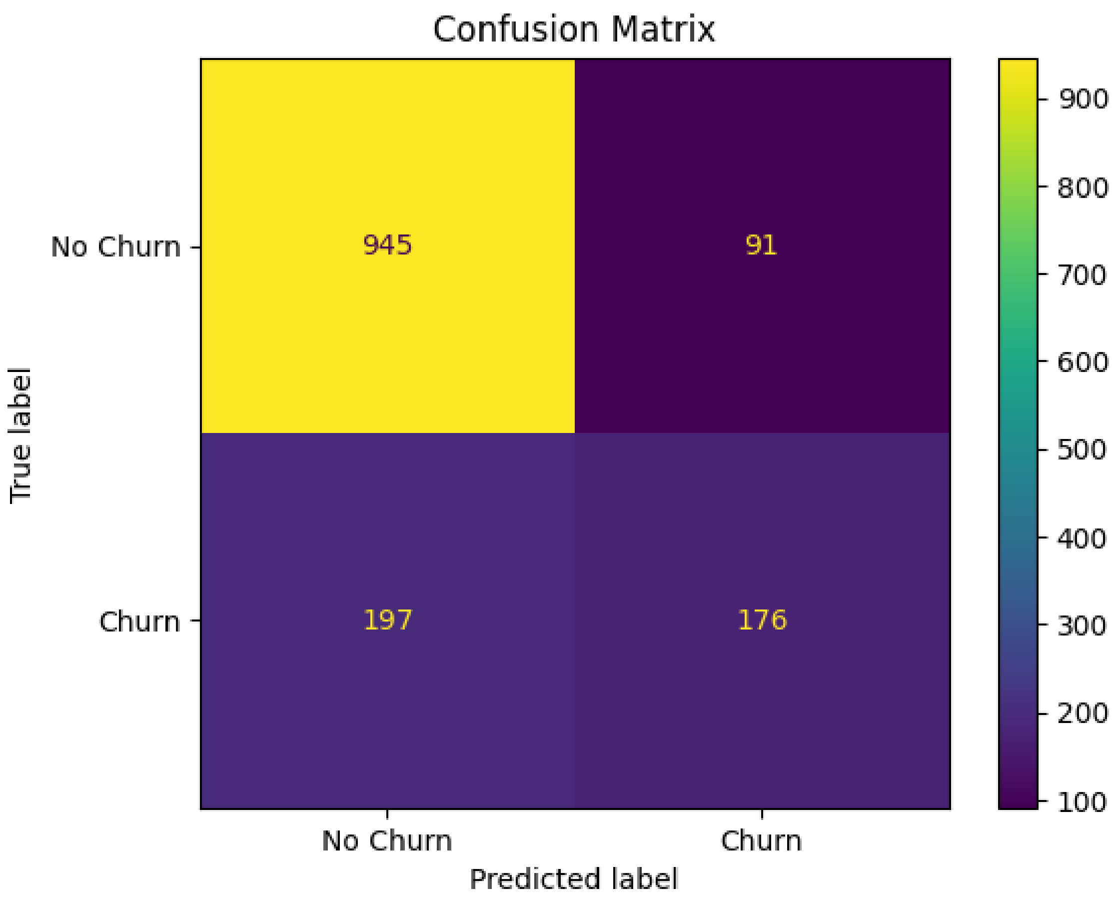
6. Model Evaluation

The model achieved an overall testing accuracy of **79.56%**. For the No Churn class, the model achieved 0.83 precision, 0.91 recall and 0.87 F1-score. For the Churn class, it achieved 0.66 precision, 0.47 recall and 0.55 F1-score.

The results indicate that the model performs well overall, although it identifies non-churn customers more accurately than churn customers.

7. Confusion Matrix

The confusion matrix compares the actual churn values with the predictions generated by the model. It helps identify correct and incorrect classifications and provides a better understanding of the model's performance.



The confusion matrix shows 945 correctly predicted No Churn customers and 176 correctly predicted Churn customers. The model incorrectly classified 91 non-churn customers as churn and 197 churn customers as non-churn.

8. Limitation

- The model is trained using only one publicly available telecom dataset.
- The dataset contains more non-churn customers than churn customers, resulting in class imbalance.
- The model has lower recall for churn customers, so some actual churn cases may not be identified.

9. Future Scope

- Larger and more recent telecom datasets can be used to improve prediction performance.
- Other Machine Learning algorithms and hyperparameter tuning can be used for comparison and optimization.
- The model can be integrated into a web application or telecom customer management system for practical use.

10. Conclusion

A Telecom Customer Churn Prediction Model was developed using the Random Forest algorithm. After preprocessing and training, the model achieved an accuracy of approximately **79.56%** on the testing data. The project demonstrates the practical use of Machine Learning for predicting telecom customer churn.